

TG-142

Wednesday, July 31, 2024

11:02 AM

Linac QA Procedure (Daily)		Tolerance	
	IMRT	SRS/SBRT	
Dosimetry			
X-ray output constancy	3%		
Electron output constancy	3%		
Mechanical			
Localizing Lasers	2 mm	1 mm	
Distance indicator (ODI)	2 mm	2 mm	
Collimator size indicator	2 mm	1 mm	
Safety			
Door interlock	Functional		
Door closing safety	Functional		
Audiovisual monitor	Functional		
Stereotactic interlocks	N/A	Functional	
Radiation area monitor	Functional		
Beam on indicator	Functional		

Daily Imaging

Imaging QA Procedure (Daily)		Tolerance	
	Non-SRS/SBRT	SRS/SBRT	
Planar kV and MV (EPID) Imaging			
Collision interlocks	Functional	Functional	
Positioning/Repositioning	2 mm	1 mm	
Imaging and treatment coordinate coincidence (single Gantry Angle)	2 mm	1 mm	
Cone-beam CT (kV and MV)			
Collision Interlocks	Functional	Functional	
Imaging and treatment coordinate coincidence	2 mm	1 mm	
Positioning/Repositioning	1 mm	1 mm	
MLC QA Procedure (Weekly)		Tolerance	
Qualitative (picket fence)	Visual Inspection for deviations i.e. increased interleaf transmission		
Wedge QA Procedure (Daily)		Tolerance (Dyn, Uni, Virt)	
Morning check-out run for one angle	Functional		

Monthly Linac

Linac QA Procedure (Monthly)		Tolerance	
	IMRT	SRS/SBRT	
Dosimetry			
X-ray output constancy	2%		
Electron output constancy	2%		
Backup monitor chamber constancy	2%		
Typical dose rate output constancy	2%		
Photon beam profile constancy	1%		
Electron beam profile constancy	1%		
Electron beam energy constancy	2%/2 mm		
Mechanical			
Light radiation field coincidence	2 mm or 1% on a side		
Light radiation field coincidence (asymmetric)	1 mm or 1% on a side		
Lasers vs. front pointer	1 mm		
Gantry/Collimator angle indicators	1.0 degrees		
Accessory trays (i.e. port film graticule)	2 mm		
Jaw position indicators (symmetric)	2 mm		
Jaw position indicators (asymmetric)	1 mm		
Cross hair centering (walkout)	1 mm		
Treatment couch position indicators	2 mm/ 1 degree	1 mm/ 0.5 degree	
Wedge placement accuracy	2 mm		
Compensator placement accuracy	1 mm		
Latching of wedges, blocking trays	Functional		
Localizing lasers	1 mm	1 mm	
Safety			
Laser guard interlock test	Functional		
Respiratory Gating			
Beam output constancy	2%		
Phase, amplitude beam control	Functional		
In-room respiratory monitoring system	Functional		
Gating interlock	Functional		

Monthly MLC

MLC QA Procedure (Monthly)		Tolerance	
Setting vs. radiation field for two patterns (non-IMRT)	2 mm		
Backup Diaphragm (Elekta)	2 mm		
Travel Speed (IMRT)	Loss of leaf speed >0.5 cm/s		
Leaf position accuracy (IMRT)	1 mm at four cardinal angles (use picket fence)		
Wedge QA Procedure (Monthly)		Tolerance	
	Dynamic	Universal	Virtual
Wedge factor for all energies	C.A. WF for 45 or 60 (2%)	C.A. WF for 45 or 60 (2%)	5% from unity or 2%

Monthly Imaging

Imaging QA Procedure (Monthly)		Tolerance	
	Non-SRS/SBRT	SRS/SBRT	
Planar MV imaging (EPID)			

Imaging and treatment coordinate coincidence (4 card.)	2 mm	1 mm
Scaling	2 mm	
Spatial Resolution	Baseline	
Contrast	Baseline	
Uniformity and Noise	Baseline	
Planar kV Imaging		
Imaging and treatment coordinate coincidence (4 card.)	2 mm	1 mm
Scaling	2 mm	1 mm
Spatial Resolution	Baseline	
Contrast	Baseline	
Uniformity and Noise	Baseline	
Cone Beam CT (kV and MV)		
Geometric distortion	2 mm	1 mm
Spatial resolution	Baseline	
Contrast	Baseline	
HU constancy	Baseline	
Uniformity and Noise	Baseline	

Annual Linac QA

Linac QA Procedure (Annual)		Tolerance
	IMRT	SRS/SBRT
Dosimetry		
X-ray flatness change from baseline	1%	
X-ray symmetry change from baseline	1%	
Electron flatness change from baseline	1%	
Electron symmetry change from baseline	1%	
SRS arc rotation mode (range: 0.5-10 MU/deg) (Set vs. Delivered)	N/A	MU:1 MU or 2% Gantry:1° or 2%
X-ray/electron output calibration (TG-51)	1%	
Spot check of FSCF for x-rays (two or more FSs)	2% for FS < 4x4cm 1% for FS ≥ 4x4cm	
Output factors for electron app's (check 1 app/energy)	2% from baseline	
X-ray beam quality (PDD ₁₀ or TMR ₂₀ /TMR ₁₀)	1% from baseline	
Electron beam quality (R ₅₀)	1 mm	
Physical wedge factors	2%	
X-ray monitor unit linearity	5% (2-4 MU) 2% ≥ 5 MU	
Electron monitor unit linearity	2% ≥ 5 MU	
X-ray output constancy vs. dose rate	2% from baseline	
X-ray output constancy vs gantry angle	1% from baseline	
Electron output constancy vs gantry angle	1% from baseline	
Electron and X-ray off axis constancy vs gantry angle	1% from baseline	
Arc mode (expected MU, degrees)	1% from baseline	
TBI / TSET mode	Functional	
PDD or TMR and OAF constancy (TBI / TSET)	1% TBI or 1 mm TSET from baseline	
TBI / TSET output calibration	2% from baseline	
TBI TSET accessories	2% from baseline	
Linac QA Procedure (Annual)		Tolerance
	IMRT	SRS/SBRT
Mechanical		
Collimator rotation isocenter	1 mm from baseline	
Gantry rotation isocenter	1 mm from baseline	
Couch rotation isocenter	1 mm from baseline	
Electron applicator interlocks	Functional	
Coincidence of radiation isocenter and mechanical isocenter	2 mm from baseline	1 mm from baseline
Table top sag	2 mm from baseline	
Table angle	1 degree	

Table travel maximum range movement in all directions	2 mm	
Stereotactic accessories, lockouts, etc.	N/A	Functional
Safety		
Follow manufacturer's test procedures	Functional	
Respiratory Gating		
Beam energy constancy	2%	
Temporal accuracy of phase/amplitude (gating on)	100 ms of expected	
Calibration of surrogate for respiratory phase/amplitude	100 ms of expected	
Interlock Testing	Functional	

Annual Imaging

Imaging QA Procedure (Annual)	Tolerance		
	Non-SRS/SBRT		SRS/SBRT
Planar MV Imaging (EPID)			
Full Range of travel SDD	±5 mm		
Imaging Dose	±1 cGy baseline		
Planar kV Imaging			
Imaging Quality (Spatial Resolution, Contrast), Energy	Baseline		
Imaging Dose	±1 mGy (Static Imaging) & ±1 cGy/min (fluoroscopy) of Baseline		
Cone Beam CT (kV and MV)			
Imaging Quality (Uniformity, Spatial Resolution, Contrast)	Baseline		
Imaging Dose	±1 cGy baseline		
MLC QA Procedure (Annual)			
MLC Transmission (Average of Leaf and Interleaf Transmission)	0.5% from Baseline		
Leaf Position Repeatability	1 mm		
MLC Spoke Shot	1 mm Radius		
Coincidence of Light Field and X-Ray Field (All Energies)	2 mm		
Segmental IMRT (Step & Shoot) Test	<0.35 cm Max Error RMS, 95% of Error Counts <0.35 cm		
Moving Window (4 Cardinal Angles)			
Wedge QA (Annual)			
Check of Wedge Angle for 60°, Full Field and Spot Check for Intermediate Angle, Field Size	Dynamic	Universal	Virtual
	Check of Off-Center Ratios @ 80% Field Width @ 10 cm to be Within 2%		